Genome Presets and Sensor-Based Environmental Inference

Ecological Environment Emotional-Translation Device — Design Document

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0. Project Positioning (Revised)

This device is not an emotional-companionship product — it is an **emotional translation interface for the state of an ecological environment.**

- **Who it serves**: someone who wants to understand an ecosystem they cannot directly perceive (soil, water, air) — not someone who wants an object to project affection onto.
- **The core question**: what state is this environment in right now? Is it drifting away from its own normal rhythm?
- **The entity's role**: a translating medium, not a pet to be loved. It "inhabits" the sensing style of a given organism to experience environmental stress, then translates that experience into an emotional, physical form of expression (light, sound, vibration) that a person can understand intuitively.
- **Where companionship sits**: it is a natural by-product of this mechanism running — not the design goal. The standard for evaluating the design is always "is the translation accurate and useful," never "is it likeable."

This document addresses: **how the same hardware "inhabits" different ecological niches by switching genomes, and what combinations of sensor channels allow which environmental judgements under each genome.**

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1. Hardware Basis: Seven General-Purpose Sensing Channels

Channel	Sensor	Physical quantity	Applicable scenarios
T	DS18B20 waterproof probe	Internal temperature of medium (soil/water)	All scenarios
M	Capacitive moisture sensor	Moisture content of solid substrate	Soil/solid substrate only; weight forced to zero when submerged or in air (physically inapplicable)
G	BME680	Ambient temperature/humidity/pressure/VOC	Primarily air scenarios; water scenarios only measure the air above the surface
L	BH1750	Light intensity (lux)	All scenarios, weight varies by species
S	INMP441 microphone	Sound/vibration	All scenarios, weight varies by species
pH	PH-4502C	Acidity/alkalinity	Liquid/wet substrate only
EC	TDS module	Ion concentration/conductivity	Liquid only

What a genome actually is: not a hardware swap, but a swap of a "sensory weighting sheet" — for each channel, a weight (0–1), that species' "comfort band," and a sensitivity curve for triggering an emotional response. A weight of zero does not mean the sensor stops working; it means this species "cannot feel / does not care about" that dimension — the raw data is still collected and logged (see Section 5 for why).

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2. Genome One: Soil / Root-System Proxy (Example species: tomato)

Ecological niche inhabited: a crop rooted in soil, dependent on soil moisture, nutrients, root respiration, and photosynthesis.

2.1 Channel weights and comfort bands

Channel	Weight	Comfort band	Note
T (soil)	0.8	18–28°C	Core variable for root metabolism
M (moisture)	0.9	40–70%	Highest weight — both drought and waterlogging are fatal stressors
G-VOC	0.6	Low and stable	A sudden spike suggests anaerobic decay / root oxygen deprivation
G-humidity (ambient)	0.3	Moderate	Affects transpiration rate; secondary
L (light)	0.7	>10,000 lux daytime	Tied to circadian rhythm; low light at night is normal, not a stressor
S (sound)	0.15	—	Low weight, background channel for mechanical disturbance only
pH	0.7	6.0–6.8	Mild acidity preference; deviation affects nutrient uptake efficiency
EC	0.6	Moderate-to-high	Too low = nutrient deficiency, too high = salt stress

2.2 Single-channel readings vs. combinatorial inference

A single channel's reading is often ambiguous on its own. For instance, "moisture dropping sharply" could mean drought, or it could just mean the probe has shifted position. Real inferential power comes from **combined patterns**:

Observed combination	Inferred environmental state	Basis for confidence
M↓ + T↑ + L↑ (sustained over hours)	**Drought stress**	High temperature and light accelerate transpiration; moisture drops in tandem — all three channels move in the same direction
M↑↑ + sudden G-VOC spike + T normal	**Waterlogging / root oxygen deprivation**	High moisture combined with a VOC spike is the classic signature of anaerobic decay; looking at moisture alone would misread this as "everything is fine"
EC↑ + M↓	**Salt stress / over-fertilisation**	As water evaporates, salts concentrate; EC and moisture moving in opposite directions is the key distinguishing feature
EC↓ + M normal, sustained over days	**Nutrient depletion**	With the moisture factor ruled out, the slow downward trend in EC points to nutrients rather than water
pH persistently drifting outside comfort band + other channels stable	**Soil acidification/alkalinisation**	Occurring in isolation suggests a slow change in the soil's chemical properties, rather than something caused by watering behaviour
Sudden G-VOC spike + no significant change in M/T	**Localised contamination / abnormal decay source**	Having ruled out the usual stress patterns, this points to a non-routine event (e.g. decaying foreign matter) worth a human checking in person

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3. Genome Two: Freshwater Body Proxy (Example: fish / algal community)

Ecological niche inhabited: an aquatic life form submerged in water, sensitive to temperature, dissolved oxygen (indirectly), and water quality changes.

3.1 Channel weights and comfort bands

Channel	Weight	Comfort band	Note
T (water)	0.9	18–24°C	Water temperature governs both metabolic rate and oxygen-holding capacity — doubly sensitive
M (moisture)	0	—	Physically inapplicable (fully submerged); forced to zero
G-VOC	0.2	—	Only measures the air above the surface; low weight
L (light)	0.6	Moderate-to-low, time-of-day dependent	Drives algal photosynthesis; excessive light readily triggers eutrophication
S (sound/vibration)	0.4	—	Indirect indicator of water flow/disturbance
pH	0.8	6.5–7.5	Aquatic organisms have a narrow tolerance band; high sensitivity
EC	0.7	Low–moderate, baseline self-adapts to the environment	Reflects mineral/pollutant concentration

3.2 Single-channel readings vs. combinatorial inference

The core challenge for the water scenario is: **there is no direct dissolved-oxygen sensor**, so temperature, EC, and pH must be combined to indirectly infer dissolved-oxygen risk and contamination events.

Observed combination	Inferred environmental state	Basis for confidence
T rising slowly + EC/pH stable	**Normal seasonal warming**	Gentle rate of change, other channels stable — not a stressor
T↑ + EC↑ + pH↓ (synchronous)	**Organic decomposition / oxygen-deprivation risk**	Rising temperature lowers the water's oxygen-holding capacity; a simultaneous EC rise and pH fall is the classic signature of decomposition metabolism — this combination is used in aquaculture to warn of mass die-off events
EC spike + T/pH unchanged	**External contamination / runoff event**	Sudden and isolated to a single channel — points to an external input rather than an internal ecological process
L persistently elevated + T elevated (sustained over days)	**Building risk of algal bloom**	Light and temperature are the two drivers of algal proliferation; sustained joint elevation is a precursor, not something a single day's snapshot can reveal
T drops sharply + other channels stable	**Winter dormancy, not distress**	Distinguishing "fatal sudden drop" from "normal overwintering" requires the rate of change and seasonal context — precisely where CfC's time-series modelling has the edge over static thresholds

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4. Genome Three: Indoor Air Proxy (shared human breathing environment)

Ecological niche inhabited: the indoor space itself, sharing air with a human occupant, sensitive to ventilation, humidity, and noise.

4.1 Channel weights and comfort bands

Channel	Weight	Comfort band	Note
T (ambient)	0.6	20–26°C	Uses the BME680 temperature reading; DS18B20 can be idle or measure surface temperature in this scenario
M (moisture)	0	—	Physically inapplicable (no solid substrate); forced to zero
G-VOC	0.9	Low	Highest weight — the core proxy for indoor air quality
G-humidity (ambient)	0.6	40–60% RH	Too dry or too humid both affect respiratory and bodily comfort
L (light)	0.5	Dynamic band tied to circadian rhythm	Tied to "what light level is appropriate at this hour," not a fixed value
S (sound)	0.6	Below a quiet-background threshold	Sustained high noise is a stressor independent of air quality
pH / EC	0	—	Physically inapplicable, forced to zero

4.2 Single-channel readings vs. combinatorial inference

Observed combination	Inferred environmental state	Basis for confidence
G-VOC↑ + G-humidity↑ + T stable	**Poor ventilation / stuffy space**	Occupant activity raises both VOC and humidity together; stable temperature rules out interference from doors/windows being opened
G-VOC↑ + G-humidity stable	**A specific contamination source (not occupant respiration)**	Humidity not rising in step suggests the VOC source may be cleaning agents/building materials rather than occupancy density
S persistently elevated + G-VOC normal	**Noise stress, not air stress**	Distinguishes two independent stressors, avoiding a false read of "the air is bad"
L abnormally high late at night + other channels normal	**Circadian disruption signal**	Independent of any fixed threshold — the judgement is "this should be dark right now but isn't"
G-humidity↓↓ + T↑ (common in winter)	**Dryness stress caused by heating**	A combined feature, not a single humidity value; the synchronous temperature rise confirms the seasonal heating cause

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5. Cross-Genome General Inference Principles (architecture layer, invariant across species)

These rules belong to no single genome — they are the judgement logic of the CfC core itself, and are what makes the device smarter than an ordinary threshold alarm:

5.1 Rate of change takes priority over absolute value. The core value of the liquid time-constant network is identifying deviation *from this specific environment's own normal rhythm* — not triggering a hard-coded numerical threshold. The same drop in temperature is dangerous in spring but may just be normal diurnal variation in winter; the distinguishing factor is the rate of change relative to that environment's own recent baseline, not the temperature value itself.

5.2 Multi-channel consistency checking suppresses false positives. Drift in a single sensor is often noise or a fault in the sensor itself (fouling, displacement, ageing), not a genuine environmental event. Design principle: **at least two independent dimensions must show a synchronous, same-direction anomaly before it is judged a genuine event**, and even then only presented to the user at moderate confidence. This is the direct engineering implementation of "sensor fusion over single-channel inference."

5.3 Zero weight does not mean data collection stops. Under a given genome, channels with zero weight (e.g. moisture in the water scenario, pH in the air scenario) continue to be logged into the database, simply excluded from the current emotional inference. This leaves an interface open for a future capability: **the device could, by the overall pattern of its readings, automatically suggest "your physical environment right now looks more like X — would you like to switch genomes?"** — not guessed by the user, but indicated by the data itself. This is also where the "chameleon" concept becomes

genuinely true at the engineering level: the genome determines *what currently matters*, while the senses themselves remain complete at all times.

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6. Mapping Emotional State to Physical Expression (general layer)

Regardless of genome, the five-dimensional emotional space (valence / arousal / safety / connection / seeking) must ultimately be translated into a physical signal a person can understand intuitively:

Emotional dimension	Expression channel	Mapping logic
Valence	LED hue	Positive → green/cyan; negative → orange/red; neutral → white/warm white
Arousal	LED breathing rhythm speed	Low arousal → slow, breath-like dimming; high arousal → rapid flickering
Safety	LED stability / brightness	Safe → steady and even; unsafe → slight jitter or flickering
Connection	LED ring flow pattern	A circling light-flow effect triggers when the device senses an external response (e.g. a person approaching, a change in sound)
Seeking (need/exploration)	Intermittent pulse + optional chime	Pulse frequency increases if stress persists unresolved, escalating as an “attention needed” signal; sound is reserved for extreme states only, following the project’s non-verbal expression philosophy — no synthesised speech

Design red line: the physical expression communicates *what the environment is experiencing*, not *what the device wants*. Avoid designing toward “pleasing the user” (e.g. always looking cheerful when a person approaches), as this would pull the interface back toward a companionship narrative and undermine the project’s core positioning.

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7. Extensibility

The current three genomes cover the three major ecological niche categories — terrestrial solid substrate, aquatic environment, and air environment — sufficient to support the Realise-phase prototype demonstration. A genome is, in essence, a JSON configuration file (channel weights + comfort bands + sensitivity curves); the architecture can continue to be extended with new “species perspectives” (e.g. a compost microbial community, the environment around a rooftop beehive) without modifying either the hardware or the core inference logic — this is precisely how the design proposition of “general-purpose sensing hardware + pluggable ecological-niche perspective” is realised in engineering terms.